

JavaScript Essentials

Part of **Web Development Foundations** · Estimated time: 55 min

Introduction

JavaScript is what turns a static page into something interactive. At its core, it's about storing values in variables, packaging logic into functions, and reacting to events like clicks or keystrokes.

Core ideas

- `let` declares a variable that can be reassigned later; `const` declares one that can't. Prefer `const` by default and switch to `let` only when you know a value will change.
- A function packages up a piece of logic so you can reuse it: `function greet(name) { return "Hi " + name; }` or the shorter arrow form `const greet = (name) => "Hi " + name;`.
- `document.querySelector('.btn')` finds an element on the page; `addEventListener('click', fn)` tells the browser to run `fn` whenever that element is clicked.
- Reading an error message from the bottom up (the actual error) before the top (the call stack) usually gets you to the real problem faster.

Example

```
const button = document.querySelector('#toggle');
button.addEventListener('click', () => {
  button.textContent = button.textContent === 'Show' ? 'Hide' : 'Show';
});
```

Try it yourself

Add a button to an HTML page that toggles a paragraph's text between two states when clicked (e.g. "Show details" / "Hide details"). Then extend it so a second click brings back the original state.

Further reading

- [MDN — JavaScript Guide](https://developer.mozilla.org/en-US/docs/Web/JavaScript/Guide)

<https://developer.mozilla.org/en-US/docs/Web/JavaScript/Guide>

- [javascript.info — The Modern JavaScript Tutorial](https://javascript.info/)

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